Final Path Experiment - Results Analysis

Date: 2026-06-08 **Setup:** `fly_threshold_10km` (10 km fly/stationary threshold) **Split:** year-grouped 80/20 (train on earliest 80% of years, test on latest 20%) **k values:** 7, 14, 30 **Epochs:** max 200, patience 30

Triline LSTM and Transformer GPS errors are now reported with fly/no-fly gating as the primary metric. Ungated diagnostics remain in the CSVs with `ungated_` prefixes.

With Weather

k=7

Model	Mean km	Median km	P90 km	Fly Recall	Best Epoch
Direct MLP	19.02	4.85	59.98	–	9
Direct Transformer 4L	17.79	3.69	56.87	–	12
Triline LSTM 4L	17.33	1.41	53.57	0.615	4
Triline Transformer V2	17.44	1.51	57.60	0.658	17

Best mean-error model: Triline LSTM 4L (17.33 km).

k=14

Model	Mean km	Median km	P90 km	Fly Recall	Best Epoch
Direct MLP	20.68	5.13	66.20	–	5
Direct Transformer 4L	21.57	8.22	60.16	–	6
Triline LSTM 4L	18.02	1.57	59.06	0.679	4
Triline Transformer V2	18.31	1.62	62.05	0.684	30

Best mean-error model: Triline LSTM 4L (18.02 km).

k=30

Model	Mean km	Median km	P90 km	Fly Recall	Best Epoch
Direct MLP	19.08	5.02	58.84	–	6
Direct Transformer 4L	17.47	3.45	59.76	–	7
Triline LSTM 4L	16.57	1.96	51.81	0.764	2
Triline Transformer V2	16.83	1.70	53.04	0.670	10

Best mean-error model: Triline LSTM 4L (16.57 km).

Without Weather

k=7

Model	Mean km	Median km	P90 km	Fly Recall	Best Epoch
Direct MLP	19.21	4.47	62.18	–	11
Direct Transformer 4L	17.65	2.47	58.41	–	13
Triline LSTM 4L	17.77	1.59	63.24	0.724	9
Triline Transformer V2	17.58	1.52	62.75	0.696	23

Best mean-error model: Triline Transformer V2 (17.58 km).

k=14

Model	Mean km	Median km	P90 km	Fly Recall	Best Epoch
Direct MLP	20.63	5.14	64.67	–	8
Direct Transformer 4L	19.18	3.69	62.64	–	11
Triline LSTM 4L	18.31	1.65	62.29	0.731	8
Triline Transformer V2	18.11	1.64	56.66	0.738	28

Best mean-error model: Triline Transformer V2 (18.11 km).

k=30

Model	Mean km	Median km	P90 km	Fly Recall	Best Epoch
Direct MLP	20.84	5.75	62.48	–	4
Direct Transformer 4L	18.22	4.35	57.93	–	11
Triline LSTM 4L	16.51	1.73	50.50	0.702	7
Triline Transformer V2	16.76	1.89	49.49	0.718	15

Best mean-error model: Triline LSTM 4L (16.51 km).

Weather Impact

Weather effect measured as `mean_error_km_weather - mean_error_km_noweather` (negative = weather helps):

Model	k=7	k=14	k=30
Triline LSTM 4L	-0.44	-0.28	+0.05

Model	k=7	k=14	k=30
Triline Transformer V2	-0.14	+0.20	+0.07

Weather impact remains small and inconsistent. It helps both triline models at k=7, helps the LSTM at k=14, and is slightly worse at k=30.

Parameter Efficiency

Model	Params	Mean Error (k=30, weather)	Error per M param
Direct MLP	78K	19.08	244.9
Direct Transformer 4L	808K	17.47	21.6
Triline LSTM 4L	2161K	16.57	7.7
Triline Transformer V2	3297K	16.83	5.1

Conclusions

1. **Triline LSTM 4L remains the strongest overall mean-error model.** It has the best k=30 mean error in both weather and no-weather settings after gating.
2. **Gating materially improves typical triline error.** Median triline errors are now around 1.4-2.0 km because predicted no-fly days hold position.
3. **Transformer V2 is competitive in tail metrics.** It wins no-weather k=30 P90 and has close mean error, but does not consistently beat the LSTM.
4. **Weather effect is weak under year-grouped split.** The most reliable improvement is at shorter context; at k=30, no-weather triline models are slightly better.

Rollout Evaluation

Autoregressive rollouts use a minimum 60-day observed context and extend that context until observed displacement reaches 50 km when possible. Triline rollouts now gate predicted movement using each model selected fly threshold.

Model	With weather mean km	Without weather mean km	With - without mean km	With weather final km	Without weather final km
Direct	417.22	240.07	+177.15	603.58	343.52
LSTM	199.94	208.86	-8.93	248.82	266.35
Transformer	226.78	200.89	+25.89	316.39	257.22

Valid rollout paths: with weather 27, without weather 27. Skipped paths: with weather 7, without weather 7.

Weather improves the LSTM rollout but worsens Direct and Transformer rollouts in this evaluation.

Artifacts: - Per-step and per-path errors: `rollout/with_weather`, `rollout/without_weather` - Weather comparison CSV: `rollout/rollout_error_comparison.csv` - Representative GIFs: `rollout/with_weather/gifs`, `rollout/without_weather/gifs`